

One ChatGPT question “uses” one bottle of water?

Posted by u/PinkGreen666 • • 0 points • 29 comments

I'm sure I'm not the only one seeing this metric being thrown around everywhere right now.

“Just one question asked to ChatGPT is the equivalent of consuming one bottle of water” or countless articles about how much water AI is using.

Is this really true? It doesn't seem to make sense to me as water being used for cooling would be in a closed loop system right? So how is AI “consuming” water? I understand more cooling systems call for more water, but it's not just being poured out or “used up”. What's the lifetime of these cooling systems?

Is this metric being used in headlines everywhere really as dumb and misleading as it sounds? Why can't companies just use synthetic coolant? And is the demand for water based cooling systems really so great that it uses significantly more water than say a new city being built?

Is this just completely overblown? Or is the scale of things just far greater than I realize?

Comments

u/Rude-Proposal-9600 • 1 points • 2024-12-05 02:05 UTC

I assume the servers are water cooled with a closed loop system so I don't see how that is possible

u/kmurrda • 1 points • 2024-12-04 18:00 UTC

Per Google Gemini:

Here's a breakdown of how AI indirectly consumes water:

1. **Data Center Cooling:** Data centers, which house the servers that power AI, generate a lot of heat. To keep these servers running efficiently, they need to be cooled. This often involves using large amounts of water, either for direct cooling or for cooling the air used in cooling systems.
2. **Electricity Generation:** The electricity used to power AI comes from various sources, many of which require water. For example, hydroelectric power plants use water to generate electricity, while

thermal power plants often use water for cooling.

3. **Chip Manufacturing:** The production of semiconductors, which are essential components of AI hardware, is a water-intensive process. It involves cleaning and cooling equipment, as well as other water-dependent steps.

While AI offers numerous benefits, it's important to be aware of its environmental impact, including its water footprint. As AI continues to advance, it's crucial to develop more sustainable practices to minimize its resource consumption

u/Crystal_Bearer • 2 points • 2024-12-04 22:25 UTC

But neither cooling nor hydroelectric power generation actually consume water. Its evaporation loss is comparable to the open air, and does not add any pollution to the water.

u/ArtichokeEmergency18 • 5 points • 2024-12-04 14:39 UTC

To create 1 Xbox console takes 3,000+ gallons of water to produce.

Watching Netflix every month per person takes 10 gallons of water.

What? How? Water?

Netflix, for example, has thousands of servers that need to be cooled, replace worn out electronics (gallons of water to produce 1 ic), etc.